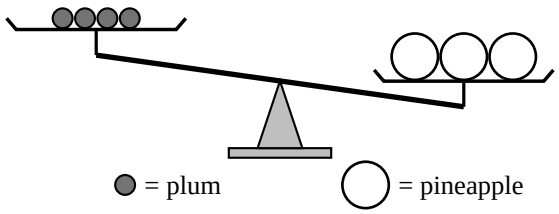
Math Kangaroo Practice

Grade 3-4 · 20 questions · standard · seed 50918

Part A: Each question is worth 3 points

1. On one plate of a balance there are 4 plums and on the other there are 3 pineapples. When we put one more pineapple, exactly like the others, on the plate with the plums, the balance is level.



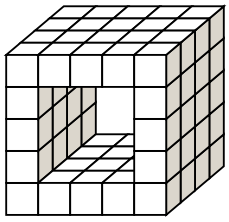
One pineapple weighs the same as how many plums?

A. 5 plums ***B.*** 4 plums ***C.*** 3 plums ***D.*** 2 plums ***E.*** 1 plum

2. Emma, Noah, Aisha and Mia each play a different instrument: the drum, the flute, the guitar and the violin. Aisha plays with sticks. Noah plays an instrument with strings. Emma plays an instrument with strings. Emma does not need a bow.
Who plays the drum?

A. Noah ***B.*** Mia ***C.*** Aisha
D. Emma ***E.*** Impossible to determine

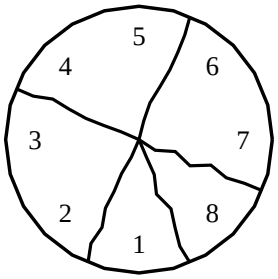
3. Mia built this tunnel out of identical cubes. The hole goes straight through from the front to the back.



How many cubes did Mia use?

A. 48 ***B.*** 64 ***C.*** 65 ***D.*** 80 ***E.*** 100

4. A round cookie with the numbers 1 to 8 around its edge is cracked into 5 pieces as shown.



Which piece has the largest sum of numbers?

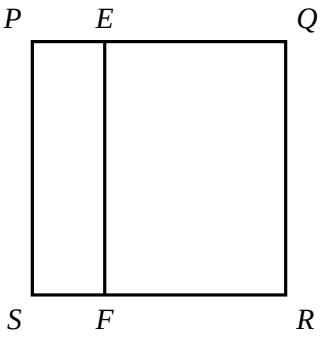
A. the piece with the 1 ***B.*** the piece with the 3 ***C.*** the piece with the 5
D. the piece with the 8 ***E.*** the piece with the 7

5. Julien, Hana, Fabienne and Leo were born in different months: October, April, June and January. Fabienne was born in winter. Julien was born in spring. Hana was born in a month that starts with J.

Who was born in January?

A. Leo ***B.*** Julien ***C.*** Fabienne
D. Hana ***E.*** Impossible to determine

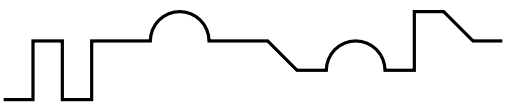
6. *PQRS* is a square. Its side is 7 cm. *PEFS* is a rectangle. Its shorter side is 2 cm.



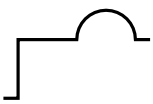
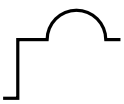
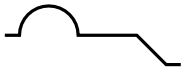
How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *PEFS*?

A. 18 cm ***B.*** 14 cm ***C.*** 10 cm ***D.*** 5 cm ***E.*** 4 cm

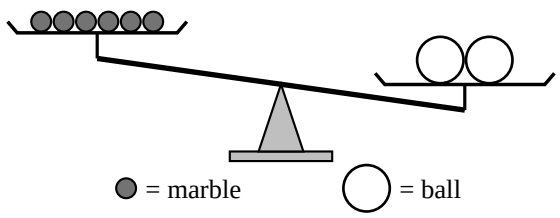
7. Lena cut a strip of paper along this line.



Which piece **cannot** come from the cut line?

A.  ***B.*** ***C.***  ***D.*** ***E.*** 

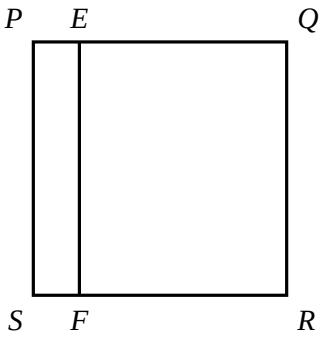
8. On one plate of a balance there are 6 marbles and on the other there are 2 balls. When we put one more ball, exactly like the others, on the plate with the marbles, the balance is level.



One ball weighs the same as how many marbles?

A. 12 marbles ***B.*** 6 marbles ***C.*** 5 marbles ***D.*** 3 marbles ***E.*** 2 marbles

9. *PQRS* is a square. Its side is 11 cm. *PEFS* is a rectangle. Its shorter side is 2 cm.

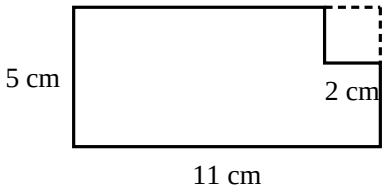


How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *PEFS*?

A. 4 cm ***B.*** 18 cm ***C.*** 22 cm ***D.*** 26 cm ***E.*** 36 cm

Part B: Each question is worth 4 points

10. A rectangle is 11 cm long and 5 cm wide. A square with side 2 cm is cut off one corner.

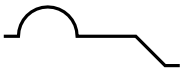
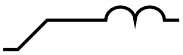
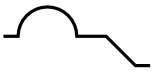
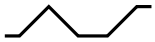


What is the perimeter of the remaining shape?

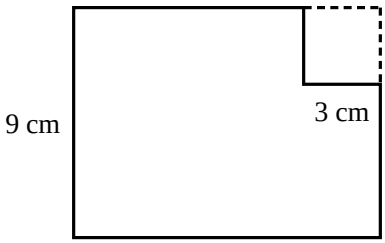
A. 24 cm ***B.*** 28 cm ***C.*** 32 cm ***D.*** 36 cm ***E.*** 40 cm

11. Far away we see the skyline of a castle.

Which of the pieces **can** belong to the skyline?

A. ***B.***  ***C.***  ***D.***  ***E.*** 

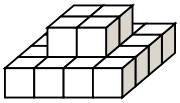
12. A rectangle is 12 cm long and 9 cm wide. A square with side 3 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 42 cm ***B.*** 43 cm ***C.*** 48 cm ***D.*** 52 cm ***E.*** 54 cm

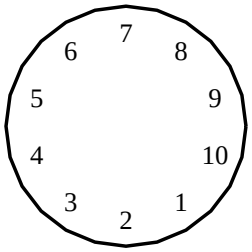
13. Tom built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 24 ***B.*** 20 ***C.*** 19 ***D.*** 18 ***E.*** 8

14. A round cookie with the numbers 1 to 10 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.



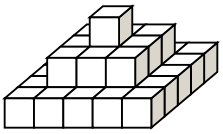
What is the smallest of these sums?

A. 18 ***B.*** 21 ***C.*** 26 ***D.*** 27 ***E.*** 28

15. Ben, Zoe, Manon and Ravi each have a different pet: a canary, a cat, a turtle and a hamster. Ravi doesn't like hamsters. Zoe owns a four-legged creature. Ben has a pet with a shell. Ravi owns a four-legged creature.
Which statement is true?

A. Ravi has a hamster ***B.*** Manon has a hamster ***C.*** Manon has a cat
D. Ben has a turtle ***E.*** Zoe has a turtle

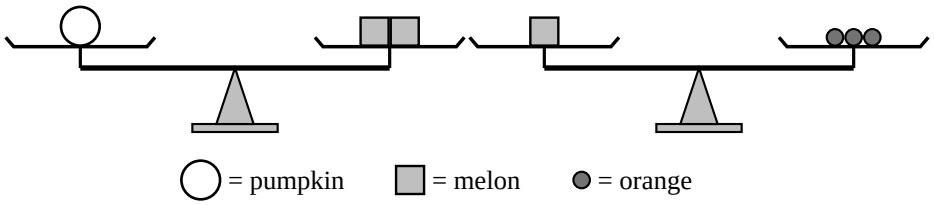
16. Sara built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 40 ***B.*** 36 ***C.*** 35 ***D.*** 34 ***E.*** 15

17. The two balances are level. One pumpkin weighs as much as 2 melons, and one melon weighs as much as 3 oranges.

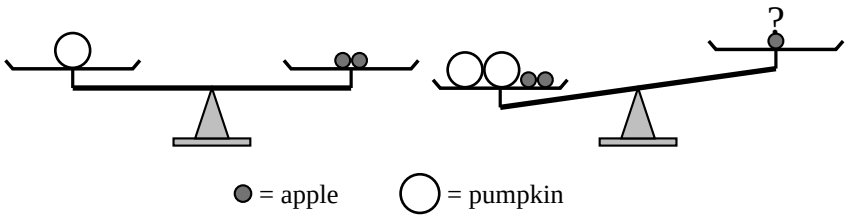


How many oranges weigh as much as 1 pumpkin?

A. 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

Part C: Each question is worth 5 points

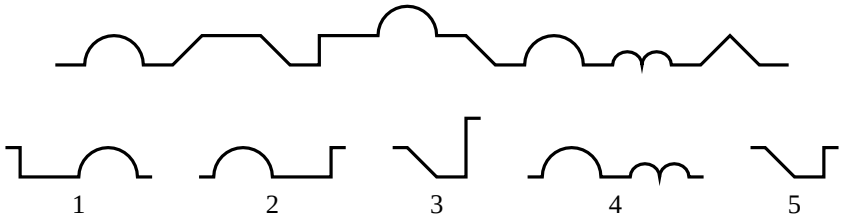
18. The first balance is level. On the second balance there are 2 pumpkins and 2 apples on the left and 1 apple on the right.



How many apples must be added to the right plate to make it level?

A. 1 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

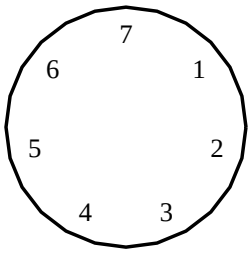
19. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

20. A wheel of fortune with the numbers 1 to 7 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 2 and 3, and between 5 and 6
B. between 3 and 4, and between 4 and 5
C. between 3 and 4, and between 5 and 6
D. between 1 and 2, and between 5 and 6
E. between 7 and 1, and between 3 and 4

Answer Key

1.	**D**	**2.**	**C**	**3.**	**B**	**4.**	**E**	**5.**	**C**	**6.**	**C**
7.	**C**	**8.**	**B**	**9.**	**B**	**10.**	**C**	**11.**	**A**	**12.**	**A**
13.	**B**	**14.**	**D**	**15.**	**D**	**16.**	**C**	**17.**	**C**	**18.**	**D**
19.	**C**	**20.**	**D**								

Solutions

1. D (2 plums) — Take 1 pineapple off both plates: 4 plums balance 3 – 1 = 2 pineapples. So one pineapple = 4 ÷ 2 = 2 plums. [2002 Q9 · one_more_balances]

2. C (Aisha) — The only possible matching: Emma – guitar; Noah – violin; Aisha – drum; Mia – flute. [2002 Q18 · who_has]

3. B (64) — The front has 5 × 5 – 3 × 3 = 16 cubes. The tunnel is 4 cubes long, so 16 × 4 = 64. [2002 Q20 · count_tunnel]

4. E (the piece with the 7) — The sums are: 6+7 = 13, 8 = 8, 1 = 1, 2+3 = 5, 4+5 = 9. [2002 Q11 · biggest_piece]

5. C (Fabienne) — The only possible matching: Julien – April; Hana – June; Fabienne – January; Leo – October. [2002 Q18 · who_has]

6. C (10 cm) — Square: 4 × 7 = 28 cm. Rectangle: 2 + 7 + 2 + 7 = 18 cm. Difference 28 – 18 = 10 cm. [2002 Q6 · square_minus_strip]

7. C — All the other pieces can be found in the line, but this one cannot: it has the right shapes but in the wrong order. [2002 Q7 · cannot_belong]

8. B (6 marbles) — Take 1 ball off both plates: 6 marbles balance 2 – 1 = 1 ball. So one ball = 6 marbles. [2002 Q9 · one_more_balances]

9. B (18 cm) — Square: 4 × 11 = 44 cm. Rectangle: 2 + 11 + 2 + 11 = 26 cm. Difference 44 – 26 = 18 cm. [2002 Q6 · square_minus_strip]

10. C (32 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (11 + 5) = 32 cm. [2002 Q6 · corner_cut]

11. A — Only this piece is found in the line. The others don't fit — B: it is a mirror image of a part of the line; C: it has two small bumps where the line has a round dome; D: its slope goes the wrong way; E: it has a pointed roof where the line has a round dome. [2002 Q7 · can_belong]

12. A (42 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (12 + 9) = 42 cm. [2002 Q6 · corner_cut]

13. B (20) — Count layer by layer, from the bottom: 4 × 4 + 2 × 2 = 20. [2002 Q20 · count_pyramid]

14. D (27) — All the numbers add up to 55. Two consecutive numbers that add up to 55: 27 + 28 = 55. The smallest is 27. [2002 Q11 · largest_sum]

15. D (Ben has a turtle) — The only possible matching: Ben – turtle; Zoe – hamster; Manon – canary; Ravi – cat. [2002 Q18 · which_true]

16. C (35) — Count layer by layer, from the bottom: 5 × 5 + 3 × 3 + 1 × 1 = 35. [2002 Q20 · count_pyramid]

17. C (6) — One pumpkin = 2 melons = 2 × 3 = 6 oranges. [2002 Q9 · chain]

18. D (5) — One pumpkin = 2 apples, so the left plate is worth 2 × 2 + 2 = 6 apples. The right plate has 1, so add 6 – 1 = 5. [2002 Q9 · what_to_add]

19. C (2) — Pieces 4, 5 fit. Not: 1 (it steps down where the line steps up); 2 (it has the right shapes but in the wrong order); 3 (its step is taller than in the line). [2002 Q7 · how_many_fit]

20. D (between 1 and 2, and between 5 and 6) — Total 28, so each piece has 14: 2+3+4+5 = 14 and 6+7+1 = 14.
[2002 Q11 · straight_crack]